

# Predictively Oriented PGD

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# Problem

**Problem** We observe data generated by an unknown distribution  $P_0$ . We choose a model  $\mathcal{M}_\Theta = \{P_\theta, \theta \in \Theta\}$  and a prior  $\Pi_\eta \in \mathcal{P}(\Theta)$ .

**Question** what if our model is wrong?

- In a perfect world  $P_0 = P_{\theta^*}$  (true if well specified).
- In practice, most models are misspecified to some extent

**Aim** construct a posterior which is more robust to misspecification

# Gaussian Bayes

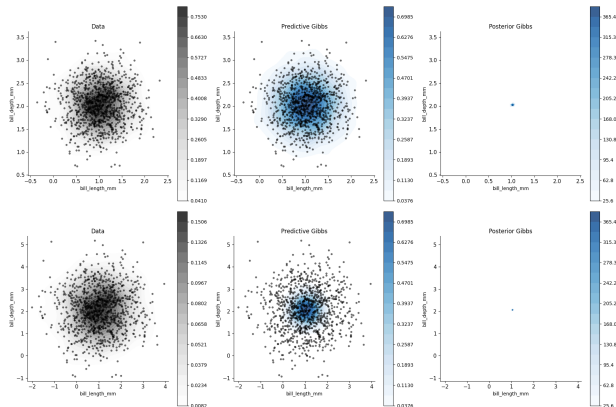


Figure: Top row well specified, bottom row misspecified. Model  $P_\theta$  is bivariate isotropic Gaussian  $\mathcal{N}(\mu, \sigma^2 \text{Id})$ .

# Bayes posteriors

When the model  $P_\theta$  has a density  $p(\cdot|\theta)$  we can write the Bayes posterior as

$$p_\eta(\theta|x_{1:M}) \propto p(x_{1:M}|\theta)\pi_\eta(\theta)$$

Optimal for well-specified models

$$p_\eta(\theta|x_{1:M}) \Rightarrow \delta_{\theta^*}$$

**Prediction:** average over posterior

$$P_q = \int_{\Theta} P_\theta q(\theta) d\theta$$

# Bayes as an optimisation problem

An equivalent way to view Bayesian inference is as the optimisation problem

$$\mathcal{F}_M^{GB}(\eta, q) = \int \log \left( \frac{q(\theta)}{\pi_\eta(\theta)} \right) q(\theta) d\theta - \sum_{i=1}^M \int \log(p(x_i|\theta)) q(\theta) d\theta$$

The Bayes posterior is the minimiser

$$p_\eta(\cdot|x_{1:M}) = \arg \min_q \mathcal{F}_M^{GB}(\eta, q)$$

optimising  $\eta$

$$\hat{\eta}_M = \arg \min_{\eta} \left\{ \min_q \mathcal{F}_M^{GB}(\eta, q) \right\}$$

# Predictively Oriented

Bayes:

$$\sum_{i=1}^M \int -\log(p(x_i|\theta))q(d\theta)$$

Predictively Oriented Posteriors (PrO):

$$\sum_{i=1}^M -\log \left( \int p(x_i|\theta)q(d\theta) \right)$$

## Remark

One can easily show  $\mathcal{F}_M^{GB}(\eta, q) \geq \mathcal{F}_M^{PrO}(\eta, q)$ .

# Optimising over probability distributions

$$\hat{\eta}, q_{\hat{\eta}}^* = \arg \min_{\eta, q} \mathcal{F}(q, \eta)$$

| Euclidean optimisation              | Wasserstein optimisation              |
|-------------------------------------|---------------------------------------|
| Variable: $\theta \in \mathbb{R}^d$ | Variable: $q \in \mathcal{P}(\Theta)$ |
| Objective: $f(\theta)$              | Objective: $\mathcal{F}(q)$           |
| Gradient descent                    | Wasserstein gradient flow             |
| One point moves                     | A cloud of particles moves            |

A particle approximation of  $q_t$  leads to stochastic dynamics.

# SDEs

For the Gibbs case:

$$d\Theta_t = \left[ \nabla_{\theta} \log p_{\eta_t}(\Theta_t) + \sum_{i=1}^M \nabla_{\theta} \log p(x_i | \Theta_t) \right] dt + \sqrt{2} dW_t \quad (2.1)$$

For the PrO case:

$$d\Theta_t = \left[ \nabla_{\theta} \log p_{\eta_t}(\Theta_t) + \sum_{i=1}^M \frac{p(x_i | \Theta_t)}{\mathbb{E}_{\Theta \sim q_t}[p(x_i | \Theta)]} \nabla_{\theta} \log p(x_i | \Theta_t) \right] dt + \sqrt{2} dW_t \quad (2.2)$$

Optimise hyperparameter

$$\partial_t \eta_t = -\nabla_{\eta} \mathcal{F}(q_t, \eta_t)$$

# Experiments

Some toy experiments. Here we considered the linear regression model

$$x_i = \theta_1 + z_i \theta_2 + \varepsilon_{x,i} \quad (2.3)$$

$$\theta = \eta + \varepsilon_\theta \quad (2.4)$$

with well specified data i.e.  $\varepsilon_{x,i} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_x^2)$ .

- 1 heavy-tailed noise  $\varepsilon_{x,i} \stackrel{iid}{\sim} t_{df}(0, \sigma^2)$  where  $df$  is degrees of freedom and can be chosen
- 2  $\theta_2 = 2 \cdot (-1)^{\xi_i}$ , with  $\xi_i \stackrel{iid}{\sim} \text{Ber}(\frac{1}{2})$
- 3 both  $\varepsilon_{x,i} \stackrel{iid}{\sim} t_{df}(0, \sigma^2)$  and  $\theta_2 = 2 \cdot (-1)^{\xi_i}$ , with  $\xi_i \stackrel{iid}{\sim} \text{Ber}(\frac{1}{2})$

# Experiments

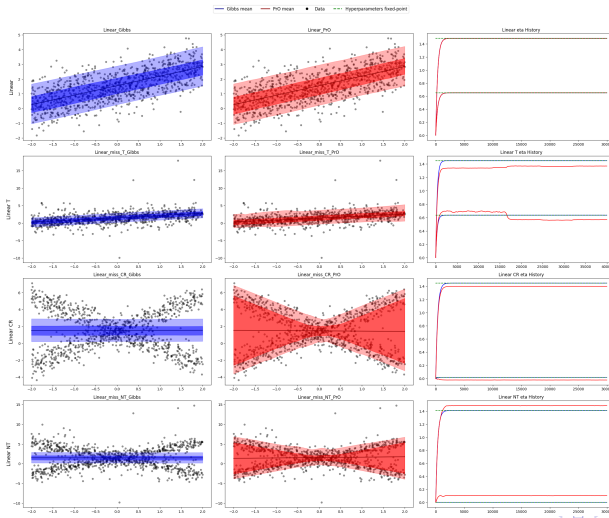


Figure: Linear regression experiments and path of hyperparameters (rightmost).

# Palmer Penguins data

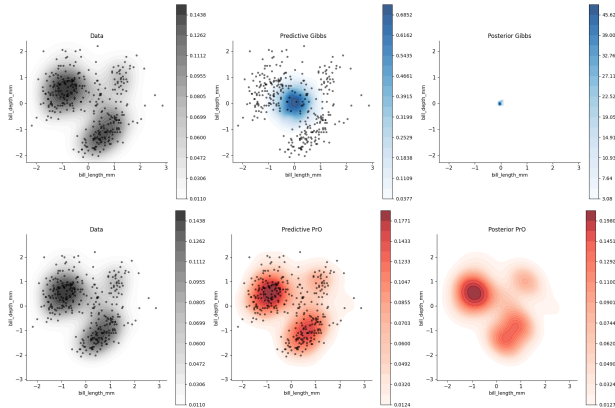


Figure:  
Palmer's  
Penguins plots:  
top row Gibbs,  
bottom row  
PrO. Take  
model  $P_\theta$  to be  
bivariate  
isotropic  
Gaussian.

# Conclusion

- Standard Bayes is optimal under correct specification.
- We propose Predictively Oriented Posteriors.
- PrO optimises predictive performance directly.
- PrO are more robust under misspecification.

# Thanks for listening